

TERRAFORM

Advanced Interview Guide

150 Questions & Answers

100 Advanced Technical Questions

50 Real-World Scenario Based Questions

150

Total Q&As

50

Scenarios

10+

Topic Areas

All

Levels

Covers: Core Concepts • Modules • State Management •

Security • CI/CD • Enterprise • Best Practices

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Part 2: Scenario-Based Questions (50 Q&As;)

Real-world troubleshooting and design scenarios

Section 1

Core Concepts

Q1 – Q10

★ *Tip: Always explain WHY, not just WHAT. Interviewers love candidates who understand the reasoning behind Terraform's design decisions.*

Q 1. What is Terraform and why is it used?**Answer:**

Terraform is an open-source Infrastructure as Code (IaC) tool by HashiCorp. It lets you define your cloud infrastructure (servers, databases, networks) in simple configuration files. Instead of clicking through a cloud console, you write code that describes what you want, and Terraform builds it automatically. It works with AWS, Azure, GCP, and 300+ other providers. The big benefit is repeatability — run the same code and get the same infrastructure every time.

Q 2. What is the difference between Terraform and CloudFormation?**Answer:**

Both are IaC tools, but CloudFormation only works with AWS, while Terraform works with any cloud (AWS, Azure, GCP, etc.) and even on-premise tools. Terraform uses HCL (HashiCorp Configuration Language) which is simpler and more readable than CloudFormation's JSON/YAML. Terraform also has a great plan command that shows you what will change before you apply it. CloudFormation is tightly integrated with AWS services and has native rollback support. Choose Terraform for multi-cloud, CloudFormation for AWS-only deep integration.

Q 3. What is a Terraform provider?**Answer:**

A provider is a plugin that lets Terraform talk to a specific platform. Think of it as a translator between Terraform and the cloud API. For example, the AWS provider knows how to create EC2 instances, S3 buckets, etc. You declare providers in your config like: `provider 'aws' { region = 'us-east-1' }`. Terraform downloads the provider plugin automatically when you run `terraform init`. Each provider is maintained either by HashiCorp, the cloud vendor, or the community.

Q 4. Explain the Terraform state file and why it is important.**Answer:**

The state file (`terraform.tfstate`) is Terraform's memory — it keeps track of what resources it created and their current settings. Without it, Terraform would not know that an EC2 instance it created last week still exists. Every time you run `plan` or `apply`, Terraform reads the state file, compares it to your config and the real cloud, then figures out what changes to make. It stores resource IDs, IPs, and metadata. Losing the state file means Terraform loses track of your infrastructure, which can lead to duplicates or orphaned resources. Always store it safely in a remote backend like S3.

Q 5. What is a remote backend and why should you use one?**Answer:**

By default, Terraform saves the state file on your local machine. A remote backend stores it in a shared location like AWS S3, HashiCorp Consul, or Terraform Cloud. This is important for teams because: (1) Everyone shares the same state, (2) State locking prevents two people from applying changes at the same time, (3) The state is backed up and versioned. A typical setup uses S3 for storage and DynamoDB for locking. Without a remote backend, two team members running terraform apply simultaneously can corrupt the state.

Q 6. What is state locking in Terraform?**Answer:**

State locking is a safety mechanism that prevents multiple terraform apply commands from running at the same time on the same state file. When you run terraform apply, Terraform places a lock on the state. If someone else tries to run apply at the same time, they get an error telling them the state is locked. This prevents race conditions where two changes could conflict and corrupt the state file. With AWS backend, DynamoDB is used for locking — Terraform creates a lock record in a DynamoDB table before modifying state.

Q 7. What are Terraform workspaces?**Answer:**

Workspaces let you have multiple independent state files from the same configuration. Imagine you want to deploy the same infrastructure to dev, staging, and production. Instead of copying your code three times, you create three workspaces: terraform workspace new dev, terraform workspace new prod. Each workspace gets its own state file. You can use terraform.workspace in your code to set different values per environment. However, for large teams, separate directories or separate repos per environment is often cleaner than workspaces.

Q 8. What is the difference between terraform plan and terraform apply?**Answer:**

terraform plan is a dry run — it shows you exactly what Terraform will create, modify, or destroy without actually doing anything. It is like a preview. terraform apply actually executes those changes. Best practice is to always run plan first, review the output, and only then run apply. In CI/CD pipelines, you might save the plan to a file (terraform plan -out=plan.tfplan) and then apply exactly that plan (terraform apply plan.tfplan) to ensure nothing changes between plan and apply.

Q 9. What is terraform import and when do you use it?**Answer:**

terraform import is used to bring existing resources (created manually or by other tools) under Terraform's management. For example, if someone manually created an S3 bucket and now you want Terraform to manage it, you run: `terraform import aws_s3_bucket.my_bucket my-bucket-name`. This adds the resource to the state file. Important: import only updates the state — you still need to write the matching Terraform config manually. From Terraform 1.5+, there is an import block that can auto-generate config for you.

Q 10. What is the terraform taint command?**Answer:**

terraform taint marks a resource as 'tainted', meaning it will be destroyed and recreated on the next terraform apply, even if no configuration changed. Use this when a resource is in a broken state (e.g., an EC2 instance that is failing health checks) and you want to force Terraform to replace it. In newer Terraform versions (1.0+), the taint command is replaced by `terraform apply -replace=aws_instance.my_instance`, which is more explicit and safer.

Section 2

Terraform Modules

Q11 – Q20

★ *Tip: Module questions are very common. Know the difference between root and child modules, and be able to explain module versioning.*

Q 11. What is a Terraform module and why use it?**Answer:**

A module is a container for multiple related resources that are used together. Think of it like a function in programming — you write it once and call it many times. For example, a VPC module might create a VPC, subnets, route tables, and internet gateway together. You call the module with module 'my_vpc' { source = './modules/vpc' } and pass inputs. This promotes code reuse, makes configs shorter, and enforces standards across your organization. HashiCorp's public Terraform Registry has hundreds of ready-made modules you can use.

Q 12. What is the difference between a root module and a child module?**Answer:**

Every Terraform configuration has a root module — it is the main directory where you run terraform commands. When you call another module from your root module (using a module block), that becomes a child module. The root module is like the main() function; child modules are the helper functions it calls. Child modules can be local (a subfolder like ./modules/vpc) or remote (from the Terraform Registry or a Git repo). Child modules receive inputs via variables and return outputs that the root module can use.

Q 13. How do you pass data between modules in Terraform?**Answer:**

You pass data between modules using input variables and outputs. A child module declares what it needs using variable blocks. The root module passes values when calling the module. The child module declares output blocks for values it wants to share back. For example, a VPC module outputs its vpc_id. The EC2 module receives that vpc_id as an input variable. In the root module, you wire them together: vpc_id = module.vpc.vpc_id. This is how modules communicate without being tightly coupled.

Q 14. What is the Terraform Registry?**Answer:**

The Terraform Registry (registry.terraform.io) is HashiCorp's public library of providers and modules. For providers, it hosts official providers (AWS, Azure, GCP) and thousands of community providers. For modules, teams publish reusable modules for common patterns like VPC, EKS cluster, RDS, etc. You reference a registry module in your code with source = 'terraform-aws-modules/vpc/aws' and a version. Using registry modules saves time and follows community best practices. You can also create a private registry in Terraform Cloud for internal modules.

Q 15. How do you version control Terraform modules?**Answer:**

You version modules using Git tags. When you publish a module, you tag the commit (e.g., v1.0.0, v1.2.3). Callers reference the version: `source = 'git::https://github.com/org/module.git?ref=v1.2.3'`. This is critical because if you update a module and callers always pull the latest, a breaking change would affect everyone immediately. Pinning to a version means callers upgrade on their schedule. For the Terraform Registry, you specify `version = '~> 3.0'` in the module block, which means any 3.x version but not 4.0.

Q 16. What is the -target flag in Terraform?**Answer:**

The `-target` flag lets you apply or plan changes to only specific resources instead of the entire configuration. For example: `terraform apply -target=aws_instance.web` only creates or updates the web EC2 instance. This is useful in emergencies when you need to fix one resource quickly, or when you have a large infrastructure and only want to test one piece. However, overusing `-target` is a bad practice because it can leave your infrastructure in an inconsistent state where some resources are updated but dependent ones are not. Use it sparingly.

Q 17. What is a data source in Terraform?**Answer:**

A data source lets you read existing infrastructure that was not created by your current Terraform config. Instead of creating a resource, you query it. For example, if an AMI ID already exists in AWS, you use: `data 'aws_ami' 'ubuntu' { most_recent = true, filters... }`. Then you reference it as `data.aws_ami.ubuntu.id`. Data sources are read-only — they never create or modify resources. They are great for referencing shared resources (like a shared VPC created by another team) without importing them into your state.

Q 18. Explain the Terraform lifecycle meta-argument.**Answer:**

The lifecycle block customizes how Terraform manages a resource. Key settings: (1) `create_before_destroy` — create the new resource before destroying the old one (good for zero-downtime updates), (2) `prevent_destroy` — blocks terraform destroy from deleting this resource (good for databases), (3) `ignore_changes` — tells Terraform to ignore changes to certain attributes (e.g., `ignore_changes = [tags]` means tag changes made outside Terraform won't cause a diff), (4) `replace_triggered_by` — replace this resource when another resource changes. These give you fine-grained control over resource behavior.

Q 19. What is the count meta-argument in Terraform?**Answer:**

`count` lets you create multiple copies of a resource with a single block. For example, `count = 3` on an `aws_instance` resource creates 3 EC2 instances. You can reference each instance by index using `count.index` (0, 1, 2). It's great for identical resources. However, `count` has a drawback: if you remove item from the middle of a list, Terraform renumbers all subsequent resources, causing them to be destroyed and recreated. For this reason, `for_each` is generally preferred over `count` when resources have distinct identities.

Q 20. What is for_each and how does it differ from count?**Answer:**

`for_each` creates multiple resources like `count`, but uses a map or set instead of a number. Each resource gets a unique key from the map. For example: `for_each = toset(['web', 'api', 'db'])` creates three instances named `web`, `api`, and `db`. You reference the current item with `each.key` and `each.value`. The big advantage over `count`: if you remove 'api' from the set, only the `api` instance is destroyed — `web` and `db` are untouched. With `count`, removing index 1 would also affect all higher-indexed resources. Use `for_each` when resources have meaningful names.

Section 3

State Management

Q21 – Q30

★ *Tip: State management is the #1 source of production incidents. Show you understand locking, remote backends, and state migration.*

Q 21. What happens if two people run terraform apply at the same time?**Answer:**

Without state locking, both people would read the same state, make their changes independently, and then both write their state back. The second write would overwrite the first, potentially losing changes or corrupting the state. With state locking (using DynamoDB with S3 backend, or Terraform Cloud), the second person's apply would fail with a 'state locked' error until the first apply finishes. Always use remote backends with locking enabled for team environments. If a lock gets stuck (due to a crash), you can manually release it with `terraform force-unlock`.

Q 22. How do you handle sensitive data in Terraform state?**Answer:**

Terraform state can contain sensitive data like passwords, secret keys, and private IPs in plain text. To protect it: (1) Use a remote backend with encryption — S3 with server-side encryption (AES-256 or KMS), (2) Restrict access to the state bucket using IAM policies (least privilege), (3) Enable bucket versioning so you can recover from accidental corruption, (4) Mark variables as sensitive = true to prevent them from appearing in logs, (5) Use a secrets manager (AWS Secrets Manager, Vault) to store secrets and reference them in Terraform rather than hardcoding. Never commit state files to Git.

Q 23. What is terraform state mv and when do you use it?**Answer:**

`terraform state mv` moves a resource from one name to another in the state file without destroying and recreating it. Use it when you rename a resource in your config. For example, if you rename `aws_instance.web` to `aws_instance.frontend` in your `.tf` file, Terraform would try to destroy the old one and create a new one. Instead, run: `terraform state mv aws_instance.web aws_instance.frontend` — Terraform updates the state to use the new name, and the resource is untouched. Also useful for moving resources between modules or state files.

Q 24. What is terraform state rm and what is the risk?**Answer:**

terraform state rm removes a resource from the state file without destroying the actual cloud resource. After removal, Terraform forgets the resource exists. Use it when you want to stop managing a resource with Terraform (maybe another team will take it over) without deleting it. The risk: if you accidentally remove a resource from state that you still want to manage, Terraform might try to create a duplicate on the next apply. Always double-check before running state rm. It is essentially the opposite of terraform import.

Q 25. How do you migrate Terraform state between backends?**Answer:**

To migrate state from one backend to another (e.g., local to S3): (1) Add the new backend config to your Terraform files, (2) Run terraform init, (3) Terraform detects the backend change and asks if you want to migrate state — answer yes. Terraform copies the state to the new backend. For migrating between S3 buckets, update the backend config to point to the new bucket and run terraform init again. Always back up your state file before migration. For Terraform Cloud migration, use terraform login first, then update the backend block.

Q 26. What is partial configuration in Terraform backends?**Answer:**

Partial configuration means you do not specify all backend settings in your code — you pass some at runtime. This is useful for secrets (like AWS access keys) or for using the same backend config across multiple environments. Example: backend 'S3' block with only bucket name in code; during terraform init, you pass: `-backend-config='key=prod/terraform.tfstate'`. Or you put the remaining config in a separate file and run: `terraform init -backend-config=backend.hcl`. This keeps secrets out of your source code and makes the config reusable.

Q 27. What is terraform refresh and is it still used?**Answer:**

terraform refresh was a command that updated the state file to match the actual state of real infrastructure, without planning or applying changes. It was useful when someone manually changed something in the cloud and you wanted Terraform's state to reflect reality. However, from Terraform 0.15.4+, terraform refresh is deprecated. The same behavior is now built into terraform plan and terraform apply automatically — they always read the real infrastructure state. You can use `terraform apply -refresh-only` to get the old refresh behavior without applying any changes.

Q 28. How do you handle drift in Terraform?**Answer:**

Drift occurs when the real infrastructure differs from what Terraform's state or config describes (e.g., someone manually changed a security group). To detect drift: run `terraform plan` — it will show differences. To fix drift: (1) Update your Terraform config to match the manual change (adopt the drift), then run `apply`, (2) Run `terraform apply` to revert the manual change back to what Terraform expects, (3) Use `terraform apply -refresh-only` to update state to match reality without changing config. Drift prevention: use IAM policies to block manual changes and enforce all changes through Terraform.

Q 29. What is the `depends_on` meta-argument?**Answer:**

`depends_on` explicitly tells Terraform that one resource depends on another, even when there is no direct reference between them. Normally, Terraform figures out dependencies automatically by looking at resource references. But sometimes there is a hidden dependency — for example, an application needs a database to exist before it starts, but Terraform cannot see this relationship from the code. In that case, you add `depends_on = [aws_db_instance.main]` to the app resource. Use `depends_on` sparingly — only when Terraform cannot auto-detect the dependency, because it can slow down parallel execution.

Q 30. What is the difference between `terraform destroy` and `terraform apply -destroy`?**Answer:**

Both destroy all resources managed by the current Terraform config, but they differ in workflow. `terraform destroy` is the traditional command — it runs a plan showing what will be destroyed and asks for confirmation. `terraform apply -destroy` does the same thing but goes through the standard apply workflow, which is useful in CI/CD pipelines where `apply` is the standard command. In Terraform 1.0+, `apply -destroy` is preferred because you can save the destroy plan to a file (`terraform plan -destroy -out=destroy.tfplan`) and apply it later, ensuring the exact same resources are destroyed.

Section 4

Advanced Features

Q31 – Q40

★ *Tip: Dynamic blocks, lifecycle rules, and provisioners separate intermediate from advanced practitioners. Know their trade-offs.*

Q 31. How does Terraform handle resource deletion ordering?**Answer:**

Terraform automatically determines the correct order to create and delete resources based on their dependencies. When destroying, it reverses the creation order — resources that depend on others are deleted first. For example, if an EC2 instance is in a VPC, Terraform deletes the EC2 instance before the VPC. This is figured out by building a dependency graph from resource references. If you have explicit dependencies via `depends_on`, those also affect deletion order. Sometimes a resource deletion fails because a dependency is not handled correctly — you may need to manually delete the dependent resource first.

Q 32. What are Terraform locals and when should you use them?**Answer:**

locals are like local variables in a programming language. They let you compute a value once and reuse it multiple times in your config without repeating the logic. Example: `locals { common_tags = { Environment = var.environment, Project = var.project_name } }`. Then use `local.common_tags` everywhere instead of repeating the map. Use locals for: (1) Naming conventions (combining project + environment into a prefix), (2) Complex expressions you want to compute once, (3) Making your config DRY (Don't Repeat Yourself). Unlike variables, locals cannot be overridden from outside — they are purely internal.

Q 33. What is the dynamic block in Terraform?**Answer:**

The dynamic block lets you generate repeated nested configuration blocks dynamically, based on a list or map. For example, an AWS security group can have many ingress rules. Instead of writing each ingress block manually, you use a dynamic block with `for_each` over a list of ports. The dynamic block generates one ingress block per port. This is powerful for creating resources whose configuration varies by environment. It is like a for loop for nested blocks. Use it when you have a variable number of nested blocks — but avoid overusing it as it can make configs hard to read.

Q 34. What is the templatefile function in Terraform?**Answer:**

templatefile reads a file and renders it as a template, substituting variables into the text. For example, you have a `user_data.sh.tftpl` script with `${database_url}` and `${app_port}` placeholders. In Terraform, you call: `templatefile('user_data.sh.tftpl', { database_url = local.db_url, app_port = 8080 })`. Terraform replaces the placeholders and returns the rendered script. It is commonly used for EC2 user data scripts, Kubernetes manifests, or any file that needs dynamic content. The `.tftpl` extension is the convention for template files, though any extension works.

Q 35. What is the purpose of the null_resource in Terraform?**Answer:**

`null_resource` is a special resource that does nothing by itself — it has no associated cloud resource. Its power comes from using it with the `local-exec` or `remote-exec` provisioners to run scripts or commands as part of your Terraform workflow. Use cases: (1) Running a script after infrastructure is created (e.g., bootstrapping a database), (2) Triggering re-execution when something changes using the `triggers` argument, (3) Ordering — making something depend on a group of resources without tying it to one specific resource. In modern Terraform, the `terraform_data` resource replaces `null_resource`.

Q 36. Explain the Terraform graph command.**Answer:**

`terraform graph` generates a visual representation of the dependency graph between all resources in your configuration. Run it and pipe to Graphviz: `terraform graph | dot -Tsvg > graph.svg`. The output shows which resources depend on which, with arrows showing relationships. It is useful for debugging — when Terraform is applying in an unexpected order or when you want to understand a complex configuration. Each node is a resource or data source; edges show dependencies. In large configurations, the graph can be overwhelming, but it is invaluable for understanding the structure.

Q 37. What are Terraform provisioners and when should you avoid them?**Answer:**

Provisioners run scripts on local or remote machines as part of Terraform resource creation. `local-exec` runs a command on the machine running Terraform; `remote-exec` SSH into a remote machine and runs commands. They are useful for bootstrapping (e.g., installing software on a new EC2 instance). However, HashiCorp recommends avoiding provisioners because: (1) They create dependencies on connectivity and credentials, (2) They are not idempotent — if they fail halfway, Terraform has no way to know what was done, (3) Better alternatives exist: `cloud-init` for EC2 user data, or configuration management tools like Ansible or Chef called separately.

Q 38. What is Terraform Cloud and how does it differ from open-source Terraform?**Answer:**

Terraform Cloud is HashiCorp's managed SaaS platform for running Terraform. Key features beyond open-source: (1) Remote state storage and locking built-in, (2) Remote execution — plans and applies run in HashiCorp's cloud, not on your laptop, (3) Policy as Code — Sentinel policies enforce compliance rules, (4) Private module registry for your organization, (5) Audit logging, SSO, and team access controls. The free tier covers basic remote state and runs. Teams plan adds Sentinel and SSO. Open-source Terraform is the CLI tool; Terraform Cloud is the collaborative platform around it.

Q 39. What is Sentinel in Terraform?**Answer:**

Sentinel is HashiCorp's Policy as Code framework, available in Terraform Cloud/Enterprise. It lets you write policies in the Sentinel language that enforce rules on Terraform plans before they are applied. Examples: 'All EC2 instances must have a specific tag', 'No security groups may allow 0.0.0.0/0 on port 22', 'Only approved AMIs may be used'. Policies can be advisory (warn but allow), soft mandatory (can be overridden with a comment), or mandatory (block apply entirely). This ensures compliance and security standards are enforced automatically, not just by convention.

Q 40. What is the terraform_remote_state data source?**Answer:**

terraform_remote_state lets one Terraform configuration read outputs from another Terraform configuration's state. This is how different teams share infrastructure. For example, the networking team creates a VPC and exports vpc_id as an output. The application team uses:

```
data 'terraform_remote_state' 'networking' {
  backend='s3',
  config={bucket='...',
  key='networking/terraform.tfstate'}
}
```

Then they reference data.terraform_remote_state.networking.outputs.vpc_id. This decouples teams while allowing sharing of key values. The downside: it creates a tight coupling to another team's state structure.

Section 5

Security & Best Practices

Q41 – Q50

★ *Tip: Security questions test your production readiness. Always mention least privilege, secret management, and audit trails.*

Q 41. How does Terraform handle circular dependencies?**Answer:**

Terraform does not allow circular dependencies — if resource A depends on B, and B depends on A, Terraform will throw an error: 'Error: Cycle: ...' and list the cycle. To fix circular dependencies: (1) Refactor your design — usually one resource should be created first independently, (2) Break the cycle by extracting a shared resource that both depend on but that depends on neither, (3) Use `depends_on` carefully to establish order without creating cycles, (4) Sometimes the cycle appears because you referenced an attribute that creates an implicit dependency — check if you can use a different attribute or a data source instead.

Q 42. What is the moved block in Terraform?**Answer:**

The moved block (introduced in Terraform 1.1) allows you to refactor your configuration without destroying and recreating resources. If you rename a resource or move it into a module, you add a moved block telling Terraform the old address and the new address. Terraform updates the state automatically. Before moved blocks, you had to run `terraform state mv` manually. The advantage of moved blocks: they are code — they are version controlled, reviewed in PRs, and applied automatically. They are great for module refactoring where a resource moves from root to a module or between modules.

Q 43. How do you manage secrets in Terraform?**Answer:**

Secrets management in Terraform requires multiple layers: (1) Never hardcode secrets in `.tf` files — use environment variables (`TF_VAR_password`) or use a secrets manager, (2) Use AWS Secrets Manager or HashiCorp Vault as a data source to fetch secrets at runtime rather than storing them in Terraform, (3) Mark sensitive variables with `sensitive=true` so they are redacted in logs and plan output, (4) Encrypt your remote state with KMS or equivalent, (5) Use short-lived credentials (IAM roles, OIDC) instead of long-lived access keys. The goal is that secrets never exist as plain text in code, state, or logs.

Q 44. What is OIDC authentication and how does Terraform use it?**Answer:**

OIDC (OpenID Connect) allows GitHub Actions, GitLab CI, or other CI systems to authenticate with AWS (or other clouds) without storing long-lived access keys as secrets. The CI system gets a short-lived JWT token from its identity provider, and AWS IAM exchanges it for temporary credentials. In Terraform CI/CD, you configure an IAM role with a trust policy that trusts your GitHub Actions workflow. The workflow assumes this role using OIDC — no `AWS_ACCESS_KEY_ID` or `AWS_SECRET_ACCESS_KEY` needed. This is far more secure than rotating static keys, and credentials automatically expire after the job finishes.

Q 45. What is the principle of least privilege in Terraform?**Answer:**

Least privilege means giving Terraform (and its IAM role) only the permissions it needs — no more. Instead of giving Terraform an admin IAM role, create a custom IAM policy with exactly the actions it needs: `ec2:CreateInstance`, `ec2:DescribeInstances`, `s3:CreateBucket`, etc. This limits blast radius if credentials are compromised. For module development, you might create separate IAM roles with narrower permissions per team or environment. Use IAM Access Analyzer to review what permissions Terraform is actually using and trim the policy. Principle of least privilege is a security must for any production Terraform setup.

Q 46. How do you implement Terraform code review best practices?**Answer:**

Code review for Terraform includes: (1) Always require PR review before merging — no direct pushes to main, (2) Run `terraform fmt` to standardize formatting and `terraform validate` to catch syntax errors in CI, (3) Post the terraform plan output as a PR comment (using tools like Atlantis or GitHub Actions) so reviewers can see exact changes, (4) Use `terraform-docs` to auto-generate module documentation, (5) Run static analysis tools like Checkov, tfint, or terrascan to catch security misconfigurations, (6) Enforce naming conventions and tagging standards using pre-commit hooks or Sentinel policies.

Q 47. What is Terragrunt and why do teams use it?**Answer:**

Terragrunt is a thin wrapper around Terraform by Gruntwork. It solves common pain points: (1) DRY backend config — define the S3 backend once and Terragrunt generates it for each module, eliminating copy-paste, (2) Dependency management between Terraform modules with dependency blocks, (3) Running across multiple modules in order with run-all apply, (4) Keeping Terraform version pinned per project, (5) Generating providers and backend configs from a root terragrunt.hcl. Teams use Terragrunt when they have many Terraform modules and environments and want to eliminate repetition without making their Terraform code itself more complex.

Q 48. What is Atlantis and how does it improve Terraform workflows?**Answer:**

Atlantis is an open-source tool that automates Terraform via pull requests. When you open a PR with Terraform changes, Atlantis automatically runs terraform plan and posts the output as a PR comment. Team members can review the exact plan. When the PR is approved, someone comments atlantis apply and Atlantis runs terraform apply. This creates a workflow where: (1) Plans are visible to everyone in the PR, (2) Apply is gated by code review approval, (3) No one needs Terraform credentials on their laptop — Atlantis runs in a central server, (4) Complete audit trail in PR comments. It is the open-source alternative to Terraform Cloud.

Q 49. What is the difference between terraform validate and terraform plan?**Answer:**

terraform validate checks that your configuration files are syntactically correct and internally consistent — it catches typos, missing required arguments, and invalid attribute types. It is fast and works offline (no cloud API calls). terraform plan does everything validate does, plus it calls the cloud API to check what actually exists, reads the current state, and computes the full diff of what would be created/modified/destroyed. Plan is slower but gives you the full picture. In CI, run validate first as a fast sanity check, then run plan for the full preview.

Q 50. What are pre-commit hooks in Terraform?**Answer:**

Pre-commit hooks are scripts that run automatically before each git commit, catching issues before they enter version control. For Terraform, the pre-commit-terraform tool provides hooks for: terraform fmt (format files), terraform validate (validate syntax), terraform docs (update README with variable/output docs), tfllint (lint for best practices), and checkov/tfsec (security scanning). Install pre-commit, create a .pre-commit-config.yaml, and run pre-commit install. Now every commit is automatically checked. This catches formatting issues and security problems immediately, long before CI runs.

Section 6

Advanced Concepts

Q51 – Q60

★ *Tip: New features like moved blocks, check blocks, and the test framework show you keep up with Terraform's rapid evolution.*

Q 51. What is tflint and what does it check?**Answer:**

tflint is a linter for Terraform that catches issues terraform validate misses: (1) AWS-specific: invalid instance types, deprecated AMIs, invalid region names, (2) Best practices: missing required tags, variable naming conventions, (3) Possible errors: type mismatches that Terraform might handle but are probably mistakes. tflint has provider-specific plugins (tflint-ruleset-aws, azure, google) that know about each provider's valid values. It is much faster than running terraform plan for catching obvious mistakes. Integrate it in CI alongside terraform validate for a fast feedback loop.

Q 52. How do you structure a large Terraform project?**Answer:**

For large projects, the common pattern is: (1) Separate state per environment (dev/staging/prod) and per component (networking/compute/databases) to limit blast radius, (2) Use a directory structure like environments/prod/networking, environments/prod/compute — each is a separate Terraform root module with its own state, (3) Share common modules in a modules/ directory or a separate repo, (4) Use terraform_remote_state to share outputs between components, (5) Consider tools like Terragrunt to reduce backend config repetition. The key principle: small blast radius — a mistake in compute should not risk destroying networking.

Q 53. What is a backend configuration in Terraform?**Answer:**

The backend configuration in Terraform defines where and how the state file is stored and how operations are performed. The default backend is local (state on disk). Common remote backends: S3 (state in S3 bucket, locking via DynamoDB), Terraform Cloud (state + remote runs in HashiCorp Cloud), Azure Blob Storage, Google Cloud Storage, and Consul. The backend block goes inside the terraform {} block: terraform { backend 'S3' { bucket='...' key='...' region='...' } }. Backend configuration cannot use variables or references — all values must be literals or passed via -backend-config flag.

Q 54. What is the purpose of the `required_providers` block?**Answer:**

`required_providers` pins which providers your configuration uses and what versions are acceptable. Example: `required_providers { aws = { source = 'hashicorp/aws', version = '~> 5.0' } }`. The source tells Terraform where to download the provider. The version constraint (`~> 5.0` means `>= 5.0, < 6.0`) prevents unexpected breaking changes from new major versions. Always specify `required_providers` to make your code reproducible. Without it, `terraform init` downloads the latest version, which might break your code when a new major version is released. Commit the `.terraform.lock.hcl` file to lock to exact versions.

Q 55. What is `.terraform.lock.hcl`?**Answer:**

`.terraform.lock.hcl` is a lock file that records the exact provider versions and their checksums that were selected during `terraform init`. It is like `package-lock.json` in Node.js. By committing this file to Git, all team members and CI pipelines use the exact same provider versions, ensuring reproducibility. When you want to upgrade a provider, run `terraform init -upgrade`, which updates the lock file. Never manually edit the lock file. If CI fails with a checksum mismatch, the lock file might be stale — run `terraform init` and commit the updated lock file.

Q 56. How does Terraform handle provider versioning and upgrades?**Answer:**

Terraform uses semantic versioning constraints. Common constraint operators: `= 5.0.0` (exact version), `~> 5.0` (pessimistic, allows 5.x but not 6.0), `>= 5.0, < 6.0` (range), `>= 5.0` (minimum). To upgrade providers: (1) Update the version constraint in `required_providers`, (2) Run `terraform init -upgrade` to download the new version and update `.terraform.lock.hcl`, (3) Run `terraform plan` to see if any resource changes are needed (sometimes provider upgrades require config changes), (4) Test in a non-production environment first. Major version upgrades often include breaking changes documented in the provider's CHANGELOG.

Q 57. What are ephemeral resources in Terraform?**Answer:**

Ephemeral resources (introduced in Terraform 1.10) are resources whose values are obtained during a plan or apply but are NOT saved to state. They are perfect for temporary credentials, OTP tokens, or any short-lived value that should not persist. Example: fetching a Vault token that expires in an hour. If it were saved to state, the next plan would try to use a stale token. Ephemeral resources are re-evaluated each time Terraform runs. They are declared like regular resources but with `ephemeral = true` or using special ephemeral block types provided by supporting providers.

Q 58. What is the check block in Terraform?**Answer:**

The check block (introduced in Terraform 1.5) lets you write custom validation conditions that run after terraform apply and report warnings without blocking the apply. Unlike preconditions (which block apply if they fail), check blocks are non-blocking health checks. Example: after creating a web server, check that the URL returns a 200 status code. If the URL is down, Terraform shows a warning but the apply still succeeds. This is useful for smoke tests and validating that your infrastructure is actually working after creation, not just that the API calls succeeded.

Q 59. What is a precondition and postcondition in Terraform?**Answer:**

Preconditions and postconditions are lifecycle validations on resources and outputs. A precondition checks that certain conditions are true BEFORE Terraform creates/updates a resource — if it fails, Terraform stops with an error. For example, `precondition { condition = var.instance_type != 't2.micro', error_message = 'Use at least t3.micro for production' }`. A postcondition checks conditions AFTER the resource is created — if it fails, the apply fails. Example: after creating an RDS instance, postcondition that the endpoint is not empty. These provide self-documenting constraints directly in your infrastructure code.

Q 60. How do you test Terraform code?**Answer:**

Terraform testing approaches: (1) terraform validate + tflint — fast syntax and lint checks, (2) Unit testing with Terratest (Go library) or pytest-terraform (Python) — deploy real infrastructure in a test account and verify it, (3) Contract testing — use check blocks and postconditions for built-in assertions, (4) terraform test command (1.6+) — native testing framework where you write .tftest.hcl files with setup, assertions, and teardown, (5) Integration tests with Kitchen-Terraform, (6) Policy testing with Sentinel mock data. The native terraform test is the simplest starting point — no external tools needed.

Section 7

Networking & Provider Deep Dives

Q61 – Q70

★ *Tip: Multi-region and multi-account are enterprise topics. Show you understand provider aliasing and cross-account IAM roles.*

Q 61. What is the terraform test command?**Answer:**

The terraform test command (introduced in Terraform 1.6) is a built-in testing framework. You write test files ending in `.tftest.hcl`. A test file contains run blocks that call terraform apply with specific variable values, then assert that outputs match expected values. Terraform creates real resources, runs assertions, then destroys everything. You can also mock providers to avoid creating real cloud resources for unit tests. This is much simpler than Terratest because you write HCL instead of Go. Tests are version-controlled alongside your module, making it easy to verify modules work correctly after changes.

Q 62. What is provider aliasing in Terraform?**Answer:**

Provider aliasing lets you use the same provider multiple times with different configurations — most commonly for deploying to multiple AWS regions or accounts. You define multiple aws provider blocks: one is the default, others have `alias = 'us-west-2'`. Resources and modules can reference a specific alias with `provider = aws.us-west-2`. Example use: deploying a CloudFront distribution (which requires us-east-1 for ACM certificates) while your main resources are in eu-west-1. You need two AWS providers: one for each region. Aliased providers are also used for multi-account setups where different resources go to different AWS accounts.

Q 63. What is the terraform console command?**Answer:**

terraform console opens an interactive REPL (Read-Eval-Print Loop) where you can evaluate Terraform expressions and functions interactively. It is like a calculator for Terraform syntax. Use it to: (1) Test string interpolation and functions before using them in code: `cidrsubnet('10.0.0.0/16', 8, 1)`, (2) Inspect values from your current state: `aws_instance.web.public_ip`, (3) Test complex expressions without modifying your config. It reads your current state and configuration, so variables have their actual values. It is extremely useful for learning Terraform functions and debugging complex expressions.

Q 64. How do you manage multiple AWS accounts with Terraform?**Answer:**

Multi-account management uses `assume_role` in the AWS provider. Each provider block specifies a role to assume in the target account: `provider 'aws' { assume_role { role_arn = 'arn:aws:iam::ACCOUNT_ID:role/TerraformRole' } }`. You run Terraform with credentials from a central 'management' account that has permission to assume roles in other accounts. Alternatively, use multiple provider aliases, one per account. For large organizations, AWS Organizations + Control Tower structures accounts, and Terraform deploys baseline infrastructure (GuardDuty, Config, CloudTrail) to all accounts uniformly. Terragrunt's `generate` blocks make multi-account provider config less repetitive.

Q 65. How do you implement blue-green deployments with Terraform?**Answer:**

Blue-green deployments with Terraform: (1) `create_before_destroy` lifecycle — Terraform creates the new resource before destroying the old one, (2) Weighted routing — create two sets of servers (blue and green) and use AWS Route 53 weighted records or an ALB target group to split traffic. Gradually shift weight from blue to green, (3) Launch Templates with Auto Scaling Groups — deploy new launch template version, do a rolling instance refresh, then delete old template. The challenge with Terraform is that it manages state, not traffic shifts — you may need external scripts or Step Functions to gradually shift traffic between the blue and green environments.

Q 66. What is Terraform CDK (CDKTF)?**Answer:**

The Cloud Development Kit for Terraform (CDKTF) lets you write Terraform configurations using general-purpose programming languages: TypeScript, Python, Java, C#, or Go. Instead of HCL, you write real code with IDE autocompletion, type checking, loops, conditionals, and unit tests using standard test frameworks. CDKTF synthesizes (converts) your code into standard Terraform JSON configuration, which Terraform then applies normally. It is great for teams who are more comfortable with programming languages than HCL, or who need complex logic that is awkward in HCL. The tradeoff: more complexity and another abstraction layer to understand.

Q 67. What is the difference between local-exec and remote-exec provisioners?**Answer:**

local-exec runs a command on the machine where Terraform is running (your laptop or CI server). remote-exec SSH/WinRM into the remote resource (like an EC2 instance) and runs commands there. local-exec is simpler — no connectivity requirements — and is used for things like calling an API after resource creation, updating a local file, or triggering a configuration management run. remote-exec is used for bootstrapping a server directly. Both are last resorts — prefer cloud-init for server bootstrapping and separate CI jobs for post-deployment tasks. Both provisioners run only during resource creation (or when taint is used).

Q 68. What are the implications of using for_each with sensitive values?**Answer:**

Using sensitive values as for_each keys causes problems because Terraform must evaluate for_each keys during the plan phase to determine what resources exist. If the key is sensitive (e.g., a secret key name from Vault), Terraform may print it in plan output or error messages, exposing the secret. Also, Terraform currently cannot use sensitive values as for_each keys at all — it throws an error. Workarounds: use non-sensitive identifiers as keys (like account IDs or environment names) and pass sensitive values separately as resource arguments. This is a known limitation of how Terraform plans resource graphs.

Q 69. How do you handle terraform apply in a CI/CD pipeline?**Answer:**

Terraform in CI/CD best practices: (1) Use OIDC for cloud authentication — no static credentials, (2) On PR: run fmt, validate, tflint, plan, post plan as PR comment, (3) On merge to main: run apply automatically (or with manual approval gate), (4) Store state in remote backend (S3 + DynamoDB or Terraform Cloud), (5) Lock Terraform version using .terraform-version or tfenv, (6) Use -input=false and -auto-approve flags to avoid interactive prompts, (7) Save plan output to a file and apply that exact plan: terraform plan -out=tfplan && terraform apply tfplan. Tools: GitHub Actions, GitLab CI, Atlantis, Spacelift.

Q 70. What is the difference between output values and local values?**Answer:**

Output values (output blocks) expose values from your configuration to external consumers: the user running terraform apply sees them printed, other Terraform configurations can read them via terraform_remote_state, and Terraform Cloud displays them. They are like a module's public API. Local values (locals block) are internal to the current module — they simplify expressions and avoid repetition but are not visible outside. Think of outputs as public and locals as private. A module's vpc_id output is meant for others to use; a locals.name_prefix computation is just internal bookkeeping.

Section 8

Enterprise & Scale

Q71 – Q80

★ *Tip: Enterprise Terraform involves Terraform Cloud, Sentinel, and scale. Mention tools like Atlantis and Terragrunt for bonus points.*

Q 71. What is the jsonencode function and when is it used?**Answer:**

jsonencode converts a Terraform value (map, list, etc.) into a JSON string. This is commonly used when passing JSON to resources that expect string inputs. For example, AWS IAM policies are JSON strings. Instead of writing a raw JSON string (error-prone), you write: `jsonencode({ Version = '2012-10-17', Statement = [{ Effect = 'Allow', Action = 's3:*', Resource = '*' }] })`. Terraform validates the HCL structure and converts it cleanly to JSON. The opposite function, `jsondecode`, parses a JSON string into a Terraform value. These are essential for working with AWS IAM, bucket policies, and any other JSON-based configurations.

Q 72. What is the yamlencode/yamldecode function?**Answer:**

yamlencode converts a Terraform map/list to YAML string format. `yamldecode` parses a YAML string into a Terraform value. These are useful when working with Kubernetes manifests (which are YAML), Helm values files, or any YAML-based configuration. Example: encode a map to YAML and write it to a Kubernetes ConfigMap or a Helm values file. `yamldecode` is useful for reading YAML data sources or files. Note: `yamlencode` output may have different formatting than hand-written YAML, but it is functionally equivalent. Combined with the `file()` and `templatefile()` functions, these enable powerful config generation workflows.

Q 73. What is the replace function in Terraform?**Answer:**

The `replace` function substitutes parts of a string. `replace(string, search, replacement)` replaces all occurrences of `search` in `string` with `replacement`. It can use plain strings or regex. Example: `replace('hello-world-app', '-', '_')` returns `'hello_world_app'`. This is commonly used for creating resource names that meet different naming requirements. For example, S3 bucket names cannot have underscores, but your project name uses them — `replace` handles the conversion. The regex variant allows more complex patterns. Similar functions: `trimspace`, `lower`, `upper`, `format` — all useful for normalizing strings in Terraform.

Q 74. What is Terraform Enterprise?**Answer:**

Terraform Enterprise (TFE) is the self-hosted version of Terraform Cloud. Companies install it in their own data center or private cloud (AWS, Azure, GCP) when they cannot use SaaS due to compliance, data residency, or security requirements. It has all Terraform Cloud features — remote runs, private module registry, Sentinel policies, audit logs, SSO — plus additional enterprise features: clustering for high availability, an air-gapped installation option (no internet required), external Vault integration, and custom certificate support. It requires significant infrastructure to run and maintain, so most teams prefer Terraform Cloud unless they have strict self-hosting requirements.

Q 75. What is a policy set in Terraform Cloud?**Answer:**

A policy set in Terraform Cloud is a collection of Sentinel policies that are applied to one or more workspaces. Instead of attaching policies to each workspace individually, you create a policy set, add policies to it, and attach the set to workspaces (or globally to all workspaces). When a workspace runs, all attached policy sets are evaluated against the plan. Policy sets can be connected to a VCS repository — Terraform Cloud automatically imports policies from the repo. This makes policy management version-controlled and auditable. You can have global policy sets (all workspaces) and workspace-specific sets for targeted enforcement.

Q 76. How do you implement cost estimation in Terraform?**Answer:**

Cost estimation options: (1) Terraform Cloud/Enterprise — built-in cost estimation shows the monthly cost impact of each plan: how much current resources cost, how much new resources will cost, and the difference. Supports major cloud resources, (2) Infracost — open-source CLI tool that integrates with CI/CD. Run `infracost breakdown --path .` to see cost estimates. Post the cost diff as a PR comment so reviewers see cost impact, (3) Cloud provider cost calculators — some teams export Terraform plans and parse them to estimate costs using provider APIs. Cost estimation helps teams make informed decisions and catch accidentally expensive resources before deployment.

Q 77. What is the Terraform Registry private module registry?**Answer:**

The private module registry in Terraform Cloud/Enterprise is a versioned repository for your organization's internal Terraform modules. Instead of referencing modules from Git URLs (which requires authentication handling), you publish modules to the private registry and reference them cleanly: `source = 'app.terraform.io/my-org/vpc/aws'`. Modules in the private registry are versioned using Git tags, have auto-generated documentation, and integrate with Terraform Cloud's access controls so only authorized workspaces can use certain modules. This promotes module reuse across your organization while maintaining governance over approved modules.

Q 78. How do you handle Terraform at scale across hundreds of workspaces?**Answer:**

At scale, Terraform management requires: (1) Workspace organization by environment and component (a matrix of env x component), (2) Automated workspace creation using the `tfe_workspace` Terraform provider — yes, you manage Terraform with Terraform, (3) Global variables in Terraform Cloud for org-wide settings, (4) Policy sets for organization-wide compliance, (5) Run triggers — when networking workspace applies, automatically queue downstream compute workspace, (6) API-driven runs via Terraform Cloud API for orchestration, (7) Module registry for standardized building blocks, (8) Cost estimation and Sentinel policies to prevent runaway costs. Tools like Spacelift, env0, or Scalr add additional orchestration layers.

Q 79. What is the Terraform provider development workflow?**Answer:**

To create a custom Terraform provider: (1) Use the Terraform Plugin Framework (the modern approach, replacing Plugin SDK), (2) Implement resource CRUD operations — Create, Read, Update, Delete — using the Go SDK, (3) Define schema — the attributes users specify in their config and their types, (4) Implement data sources similarly but read-only, (5) Test with acceptance tests that create real resources, (6) Package and publish to the Terraform Registry. Custom providers are needed when you want to manage internal APIs or services not covered by existing providers. The Terraform Plugin Framework handles all the protocol details — you just implement the business logic.

Q 80. What is drift detection in Terraform Cloud?**Answer:**

Terraform Cloud's drift detection feature periodically runs terraform plan in health check mode on all workspaces (on a schedule you configure) without applying any changes. If drift is detected (the real infrastructure differs from the last applied state), Terraform Cloud alerts you. This is critical for compliance — you know within hours if someone manually changed something. Without automated drift detection, drift can go unnoticed for weeks. Alerts can be sent via email, Slack webhooks, or PagerDuty. You can then decide to revert the drift (apply the existing config) or codify the change and add it to your Terraform config.

Section 9

Miscellaneous Advanced

Q81 – Q100

★ *Tip: Functions like merge, flatten, and coalesce show deep HCL knowledge. Practice them in terraform console.*

Q 81. How do you use Terraform with Kubernetes?**Answer:**

Terraform + Kubernetes patterns: (1) Use Terraform to provision the cluster (EKS, GKE, AKS) and use the Kubernetes/Helm provider to deploy workloads, (2) The Kubernetes provider manages Kubernetes resources (Deployments, Services, ConfigMaps) as Terraform resources, (3) The Helm provider deploys Helm charts with values managed in Terraform, (4) Common pattern: separate state for infrastructure (EKS cluster) and for Kubernetes workloads — the workload state depends on the cluster state via `terraform_remote_state`, (5) For GitOps workflows, many teams use Terraform only for the cluster and Argo CD/Flux for workloads — each tool does what it is best at.

Q 82. What is the difference between resource and data source in Terraform?**Answer:**

A resource block creates, updates, and manages a cloud resource — Terraform owns it. If you delete the resource block, Terraform destroys the cloud resource. A data source block reads existing information about a resource — Terraform does not own it and cannot destroy it. Data sources are read-only lookups. Example: you use a resource to create an RDS database your app needs, but a data source to look up the company's shared Route 53 hosted zone managed by another team. Data sources are evaluated during plan and their values are available for use in resources. Both are called with different syntax: resource `'aws_s3_bucket'` vs data `'aws_s3_bucket'`.

Q 83. How does Terraform handle API rate limiting?**Answer:**

Cloud APIs have rate limits, and large Terraform configurations with many resources can hit them. Terraform handles this through: (1) Built-in retries — most providers automatically retry rate-limited requests with exponential backoff, (2) Parallelism control — by default Terraform applies up to 10 resources in parallel. Reduce with `terraform apply -parallelism=5` to reduce API call rate, (3) Provider-specific settings — some providers have explicit retry configuration, (4) Splitting large configurations — instead of one massive apply, split into smaller state files applied separately, (5) Using `-target` to apply subsets of resources. Rate limiting is most common in AWS with EC2 describe API calls.

Q 84. What is the merge function in Terraform?**Answer:**

merge combines multiple maps into a single map. If keys overlap, the last map's value wins. Example: `merge({ env='dev', team='platform' }, { env='prod', region='us-east-1' })` returns `{ env='prod', team='platform', region='us-east-1' }`. This is extremely useful for tags: define `default_tags` in locals and merge with resource-specific tags. Each resource gets all default tags plus its own, with resource-specific tags overriding defaults if there is a key collision. The merge function is one of the most-used Terraform functions in real-world configurations. Related: `tomap`, `zipmap`, `keys`, `values` functions for map manipulation.

Q 85. What is the flatten function in Terraform?**Answer:**

flatten takes a list of lists (potentially nested) and returns a single flat list. Example: `flatten([[1,2],[3,[4,5]]])` returns `[1,2,3,4,5]`. This is useful when you have a module that outputs a list of lists (e.g., each environment outputs a list of subnet IDs) and you want all subnet IDs in one list. Combined with `for` expressions: `flatten([for env in var.environments : [for subnet in env.subnets : subnet.id]])`. Without flatten, you would have a list of lists, which many resource arguments do not accept. The flatten function is a common tool for flattening nested data structures.

Q 86. What is the one function in Terraform?**Answer:**

The one function takes a list that must have exactly 0 or 1 elements and returns null (if 0 elements) or the single element (if 1 element). It is useful for converting optional singleton lists to optional values. Example: `one(data.aws_vpc.selected[*].id)` — if the data source found a VPC, return its ID; if not, return null. This pattern is common with data sources that use count (0 or 1) to make lookup optional. Without `one()`, you would have to use `length()` checks and conditional expressions. It makes code cleaner when dealing with 'optional' lookups. Note: it throws an error if the list has 2+ elements.

Q 87. What is the setproduct function in Terraform?**Answer:**

setproduct computes the Cartesian product of multiple sets or lists — every combination of elements from each input. Example: `setproduct(['us-east-1', 'us-west-2'], ['web', 'api'])` returns `['us-east-1','web'], ['us-east-1','api'], ['us-west-2','web'], ['us-west-2','api']]`. This is useful for creating all combinations of environments × regions × components. Combined with `for_each` and a `for` expression to convert to a map, you can create all combinations of resources without nested loops. It is an advanced function used in complex multi-region or multi-environment configurations.

Q 88. How do you implement multi-region deployments with Terraform?**Answer:**

Multi-region Terraform pattern: (1) Use provider aliases — one aws provider per region, (2) Reference resources with explicit provider = `aws.us-west-2`, (3) For modules, pass the provider using the providers argument: `module 'web' { providers = { aws = aws.us-east-1 } }`, (4) Use data sources to get region-specific values (AMI IDs differ by region), (5) Some teams use separate state files per region for isolation — if `us-east-1` has an issue, the plan/apply for `us-west-2` is unaffected, (6) Global resources (IAM, Route 53) only go in one provider. Multi-region setups are complex — consider if you truly need them or if a single region with multiple AZs suffices.

Q 89. What is the split and join function in Terraform?**Answer:**

split divides a string into a list by a separator: `split(',', 'a,b,c')` returns `['a','b','c']`. join combines a list into a string with a separator: `join(',', ['a','b','c'])` returns `'a,b,c'`. These are useful for transforming between string and list formats. Example: an environment variable contains a comma-separated list of CIDR blocks — split it to get a list you can use with `for_each`. Or join a list of security group IDs into a comma-separated string for a resource that expects that format. They are simple but frequently needed string manipulation functions in real Terraform configurations.

Q 90. What is resource drift and how does Terraform detect it?**Answer:**

Drift occurs when a cloud resource's actual state differs from Terraform's state file. This happens when: someone manually changes a resource in the cloud console, another tool modifies the resource, or the cloud provider automatically changes some attributes. Terraform detects drift during `terraform plan` — it calls the cloud API to read the current state of each resource and compares it to the state file. Differences appear as planned changes (+ for new attributes, ~ for changes, - for removed attributes). The fix is either to apply (revert to Terraform config) or update your config to match the drift and apply (adopt the change). Use `-refresh-only` to update state without changing config.

Q 91. What is the difference between input variables and environment variables in Terraform?**Answer:**

Input variables (`var.name`) are Terraform's typed, documented parameters. They have types, defaults, validation rules, and descriptions. They are set via `.tfvars` files, `-var` flags, or `TF_VAR_name` environment variables. They are the right way to parameterize Terraform configs. Environment variables in Terraform's context refer to the `TF_VAR_` prefix method of setting input variables — they map directly to input variables. Terraform also reads environment variables for provider configuration (`AWS_ACCESS_KEY_ID`, `AWS_REGION`) separately from input variables. Never confuse `var.region` (Terraform input variable) with the `AWS_REGION` environment variable — they are different concepts, though both affect which region resources go into.

Q 92. What is a Terraform stack (new feature)?**Answer:**

Terraform Stacks (introduced in late 2024, in Terraform Cloud) are a new higher-level abstraction for deploying the same Terraform configuration to multiple environments or regions simultaneously. A stack uses two new file types: a `.tfstack.hcl` for component definitions (what modules to deploy) and a `.tfdeploy.hcl` for deployment configurations (where and with what variables). Stacks can deploy one configuration to dev, staging, and prod environments in a single operation, with each deployment tracked independently. This simplifies multi-environment management compared to maintaining separate workspaces and running apply multiple times.

Q 93. How do you handle large Terraform configurations with long apply times?**Answer:**

Strategies for long applies: (1) Split the configuration — break one large state into multiple smaller ones (networking, compute, databases separately), each applying in under 5 minutes, (2) Use `-target` during development to apply only changed components, (3) Increase `-parallelism` (default 10) if the bottleneck is API calls, not rate limits, (4) Identify slow resources (often Route 53 or ACM certificate validation) and minimize changes to them, (5) Use `create_before_destroy` carefully — it increases apply time by adding a creation step before deletion, (6) Use terraform plan output to review before apply, catching unnecessary changes early. The real solution is always smaller blast radius through state splitting.

Q 94. What is the `toset`, `tolist`, and `tomap` function?**Answer:**

These are Terraform type conversion functions. `toset` converts a list to a set (removes duplicates, unordered): `toset(['a','b','a'])` returns `{'a','b'}`. `tolist` converts a set or tuple to a list (ordered). `tomap` converts an object to a map. These are needed because Terraform has strict types — a resource expecting a set won't accept a list. The most common use: `for_each` requires a set or map. If you have a list variable, convert it: `for_each = toset(var.availability_zones)`. `tomap` is used when you have an object (fixed known keys) and need a map (dynamic keys) — they look similar but behave differently in Terraform's type system.

Q 95. What is the `coalesce` and `coalescelist` function?**Answer:**

`coalesce` returns the first non-null, non-empty string from its arguments: `coalesce("", null, 'hello', 'world')` returns `'hello'`. Use it when you have multiple possible sources for a value and want the first one that is actually set. Example: the AMI ID might come from a variable (if explicitly set) or fall back to a data source — `coalesce(var.ami_id, data.aws_ami.default.id)`. `coalescelist` does the same for lists — returns the first non-empty list. These are useful for implementing fallback logic cleanly without `if/else` conditionals. Similar to the ternary operator (`condition ? value_if_true : value_if_false`) but for multiple fallbacks.

Q 96. What is the terraform fmt command and why is it important?**Answer:**

terraform fmt automatically formats your Terraform configuration files to a canonical style defined by HashiCorp. It adjusts indentation, spacing, and alignment of equals signs. Run it before committing code: `terraform fmt -recursive` to format all .tf files in subdirectories. In CI, run `terraform fmt -check -recursive` — it exits with a non-zero code if any files are not formatted, failing the pipeline. This enforces consistent code style across the team without debates about formatting preferences. Most IDEs integrate terraform fmt on save via the Terraform extension. Formatted code is easier to review in PRs because unintentional whitespace changes do not appear in diffs.

Q 97. What is the terraform show command?**Answer:**

terraform show displays a human-readable output of the state file or a saved plan file. Running terraform show on its own prints the current state in a readable format — all resources, their attributes, and values. Running `terraform show plan.tfplan` shows the contents of a saved plan file. With the `-json` flag, it outputs JSON, which is machine-readable and useful for automation: `terraform show -json plan.tfplan | jq '.resource_changes[] | select(.change.actions[] == "create")'` to see only creates. This is useful for auditing what is in state, debugging issues, and building automation around Terraform plans.

Q 98. How do you use the for expression in Terraform?**Answer:**

The for expression transforms collections. It has two forms: list: `[for item in var.items : item.name]` and map: `{for item in var.items : item.id => item.name}`. You can filter with if: `[for item in var.items : item.name if item.enabled]`. For expressions are used everywhere: transforming a list of objects into a list of IDs, creating a map from a list, filtering out disabled items. Example: create a map of instance name to instance type from a list of instance objects. Combined with locals, for expressions reduce the need for complex nested structures. They are Terraform's version of Python list comprehensions — concise and readable once you are used to them.

Q 99. What is the difference between terraform.tfvars and variables.tf?**Answer:**

variables.tf (or any .tf file) contains variable declarations — the definitions of what variables exist, their types, descriptions, defaults, and validation rules. It is like a function signature. terraform.tfvars contains the actual values assigned to those variables. It is automatically loaded by Terraform. You can also have environment-specific files like prod.tfvars loaded with terraform apply -var-file=prod.tfvars. The separation is important: variables.tf is checked into Git (it is just the schema), while .tfvars files may contain environment-specific or sensitive values and might be gitignored (though best practice is to avoid secrets in .tfvars and use environment variables instead).

Q 100. What is variable validation in Terraform?**Answer:**

Variable validation lets you define rules that values must pass, with custom error messages. Example:

```
variable 'environment' { type=string, validation { condition = contains(['dev','staging','prod'], var.environment), error_message = 'Environment must be dev, staging, or prod.' } }
```

. If someone passes environment='development', Terraform immediately fails with your custom message before making any API calls. This is far better than getting a cryptic cloud API error deep into an apply. You can have multiple validation blocks per variable. Common validations: checking CIDR block format, ensuring string length, verifying a value is within an allowed set, checking numeric ranges. Validation makes your modules self-documenting and safer to use.

PART 2**Scenario-Based Questions****50 Real-World Troubleshooting & Design Scenarios**

■ *These questions are based on real-world incidents and system design challenges. They test your ability to think through problems, not just recall facts. In interviews, walk through your reasoning step by step — process matters as much as the answer.*

Scenario 1. Your team runs terraform apply and gets 'Error acquiring the state lock'. What do you do?

Answer:

This means another terraform apply is running, or a previous one crashed and left the lock behind. Step 1: Check if anyone on the team is running Terraform right now. Step 2: If no one is running it, the lock is stale. Find the lock ID from the error message. Step 3: Run terraform force-unlock to release it. Step 4: Verify the state is not corrupted by running terraform plan. Step 5: Investigate why the previous apply crashed (check CI logs). Prevention: always use remote backends with locking, and set up alerts for long-running applies.

Scenario 2. A developer manually deleted an EC2 instance in the AWS Console that was created by Terraform. The next terraform plan shows it will be recreated. How do you handle this?

Answer:

Terraform detects the instance is gone and plans to create a new one. You have two options: Option A — let Terraform recreate it (run terraform apply). This is usually correct if the instance was deleted by mistake. Option B — if the deletion was intentional and you want to remove it from Terraform too, remove the resource block from your .tf file and run terraform apply. Terraform updates state without trying to create or delete. Never leave a gap between code and state — always reconcile them after a manual change.

Scenario 3. You need to rename a Terraform resource from aws_instance.web to aws_instance.frontend without destroying and recreating it. How?

Answer:

Use the moved block (Terraform 1.1+). In your .tf file, add: `moved { from = aws_instance.web, to = aws_instance.frontend }`. Then update the resource block name from 'web' to 'frontend'. Run terraform plan — it should show no changes to the actual EC2 instance, just a state rename. Run terraform apply. After a successful apply, you can delete the moved block (it is no longer needed). Alternatively, run `terraform state mv aws_instance.web aws_instance.frontend` before changing the code. The moved block approach is preferred because it is version-controlled.

Scenario 4. Your terraform apply is failing because a security group cannot be deleted — AWS says it is still in use. How do you debug and fix this?

Answer:

This usually means another resource (EC2 instance, RDS, EKS node) still has this security group attached. Debugging steps: (1) Run `aws ec2 describe-network-interfaces --filters Name=group-id,Values=sg-xxx` to find what is using it, (2) Check if Terraform is trying to delete the security group before the dependent resource — if so, add `depends_on` to the security group or the resource using it, (3) Sometimes the resource using it is outside Terraform's management — remove the dependency manually first. (4) Check for lifecycle rules: if the dependent resource has `prevent_destroy`, Terraform cannot delete it. Fix the dependency order and re-apply.

Scenario 5. You have a Terraform state file stored locally. Your team is growing and you need to move to a remote backend. How do you do this without losing state?

Answer:

Step 1: Configure the new backend in your Terraform files. Example: add `terraform { backend 'S3' { bucket='my-terraform-state', key='prod/terraform.tfstate', region='us-east-1', dynamodb_table='terraform-locks' } }`. Step 2: First create the S3 bucket and DynamoDB table (can be done with a separate Terraform config or manually). Step 3: Run `terraform init` — Terraform detects the backend change and asks: 'Do you want to copy existing state to the new backend?' Answer yes. Step 4: Verify with `terraform plan` — it should show no changes. Step 5: Delete the local `terraform.tfstate` file and commit the backend config to Git.

Scenario 6. You accidentally ran terraform destroy on your production environment. What do you do?

Answer:

This is a critical incident. Immediate steps: (1) Do NOT panic and do not run anything else yet — assess what was destroyed, (2) Check if your state backend has versioning — S3 bucket versioning lets you see the state before destroy. Note resource IDs from the old state, (3) Check cloud provider's recycle bin or backup features — some services (RDS, EC2) have snapshots or deletion protection, (4) If you have recent backups (database snapshots, AMIs, S3 versioning), begin restoration from those, (5) Run `terraform apply` to recreate infrastructure, then restore data from backups, (6) Prevention: use `prevent_destroy` lifecycle on critical resources, set up deletion protection on databases, require manual approval for production destroys.

Scenario 7. Two teams use the same S3 bucket as the Terraform backend but with different state file paths. Team A's apply is failing with permission errors. How do you investigate?

Answer:

The most common causes: (1) IAM policy for Team A's CI role does not allow `s3:PutObject` or `s3:GetObject` on the specific key path. Check the exact S3 key path Team A uses and verify IAM permissions, (2) The S3 bucket policy has explicit denies overriding the IAM policy, (3) KMS key permissions — if the bucket uses KMS encryption, Team A's role might not have `kms:Decrypt/Encrypt` permissions, (4) DynamoDB permissions for state locking — check `s3` and `dynamodb` permissions together, (5) Check CloudTrail logs for the exact API call that failed and the specific error. Fix the IAM policy to grant the minimum required permissions for Team A's key prefix.

Scenario 8. Your Terraform config creates an RDS database and an EC2 instance that connects to it. The EC2 instance is created before RDS is ready. How do you fix the ordering?

Answer:

Terraform should handle this automatically if the EC2 instance resource references the RDS endpoint (`aws_db_instance.main.endpoint`). The reference creates an implicit dependency — Terraform will not create EC2 until RDS exists. If you are not referencing the RDS in the EC2 resource config, add an explicit `depends_on = [aws_db_instance.main]` to the EC2 resource. However, even with `depends_on`, RDS might be created but not yet accepting connections when EC2 user data tries to connect. The proper solution is to handle connection retry logic in your application or use a startup script that waits for the database endpoint to respond before starting the app.

Scenario 9. You need to add a new required variable to a shared Terraform module that 50 other configurations use. How do you do this without breaking everything?

Answer:

Safe approach: (1) Add the new variable with a default value: `variable 'new_var' { type=string, default='sensible-default' }`. This is non-breaking — all existing callers work without changes, (2) Test with a few configurations first, (3) Gradually update callers to explicitly pass the value instead of relying on the default, (4) Once all callers pass the value explicitly, you can optionally remove the default to make it required. If you absolutely need it required from day one, coordinate with all teams to update their configs simultaneously, and do it in a scheduled maintenance window. Document the change in the module's CHANGELOG.

Scenario 10. Your terraform plan shows a resource will be destroyed and recreated, but you only changed a tag. Why is this happening and how do you fix it?

Answer:

Some AWS resources require replacement for any attribute change — they are 'immutable' in some ways. But tags should never cause replacement. Possible causes: (1) The tag change is on a resource that has `forces_new_resource` for some other reason — check if another attribute also changed, (2) You may have changed a key attribute (like the resource name or type) at the same time as the tag, (3) The resource might be using a deprecated attribute that triggers replacement. To investigate: run `terraform plan -out=plan.tfplan && terraform show -json plan.tfplan` to see exactly which attribute is triggering replacement. If it is truly just tags, check if `ignore_changes = [tags]` can be applied if external systems manage tags.

Scenario 11. You want to deploy the same Terraform configuration to dev, staging, and production environments. What are your options?

Answer:

Three main approaches: (1) Workspaces — one codebase, three workspace states. Use `terraform.workspace` to differentiate. Simple but state files share the same backend key prefix and workspaces can be confusing for complex configs, (2) Directory-per-environment — separate directories: `environments/dev`, `environments/staging`, `environments/prod`. Each has its own state and may have environment-specific `.tfvars` files. Most recommended for clarity and isolation, (3) Terragrunt — define config once in a module, use Terragrunt's DRY backend config to create per-environment state automatically. Best for large organizations with many environments. For most teams starting out, directory-per-environment is the clearest approach.

Scenario 12. A Terraform apply is stuck and has been running for 2 hours. What do you do?

Answer:

Steps: (1) Do NOT kill the process immediately — check what resource is being created/modified. Some AWS resources legitimately take a long time (RDS in multi-AZ, large EBS volumes, EKS cluster), (2) Check the AWS console — is the resource actually being created? Sometimes Terraform is waiting for a callback that never comes, (3) If clearly stuck, investigate: RDS creation usually takes 5-15 minutes; if it is 2 hours, something is wrong, (4) Check for circular dependencies in the cloud — sometimes a security group or route table has a circular reference outside Terraform, (5) If you need to kill it, ensure you release the state lock (`terraform force-unlock`) afterward, (6) Fix the underlying issue (often a dependency cycle or misconfiguration) and re-apply.

Scenario 13. Your company wants all EC2 instances to have specific tags. How do you enforce this with Terraform?

Answer:

Multiple enforcement layers: (1) Module-level — if teams use a shared EC2 module, bake the required tags into the module with locals and merge with user tags, (2) Default tags — use the AWS provider's default_tags block: all resources created by that provider automatically get those tags without any resource-level code, (3) Sentinel policy — write a Sentinel policy in Terraform Cloud that checks every plan and blocks apply if any EC2 instance is missing required tags, (4) AWS Config rules — as a backstop outside Terraform, AWS Config can detect untagged resources and trigger remediation. The most robust approach combines all three layers: default_tags for basics, shared module for standards, Sentinel for enforcement.

Scenario 14. You need to create 100 S3 buckets with names like bucket-001 to bucket-100. How do you do this efficiently in Terraform?

Answer:

Use count with format: count = 100 on the aws_s3_bucket resource. Name each bucket: bucket = format('bucket-%03d', count.index + 1) — this creates bucket-001, bucket-002, ..., bucket-100 with zero-padded numbers. Alternatively, if the bucket names come from a list, use for_each = toset(var.bucket_names). With count, you can reference each bucket as aws_s3_bucket.buckets[0] through [99]. However, creating 100 S3 buckets in one state file means any change to any bucket triggers a plan for all 100. Consider if you actually need 100 separate buckets or if one bucket with prefixes/folders would work better.

Scenario 15. A junior team member accidentally committed AWS access keys to the Terraform config. What do you do?

Answer:

Immediate response: (1) Rotate the keys immediately in AWS IAM — this invalidates the compromised keys. Do this first, before anything else, (2) Remove the keys from the config file and commit the fix, (3) Invalidate the Git history — force-push with the keys removed, or use git filter-branch/BFG Repo Cleaner to remove the keys from all commits. Note: if the repo is on GitHub/GitLab, they scan for exposed keys and may notify AWS, (4) Check AWS CloudTrail for any unauthorized API calls using the exposed keys, (5) Review IAM policy for the compromised keys — were they overprivileged? (6) Prevention: set up pre-commit hooks with tools like git-secrets or truffleHog, and use OIDC instead of long-lived keys.

Scenario 16. You need to refactor a Terraform configuration by moving several resources into a module. How do you do this without downtime?

Answer:

Safe refactoring steps: (1) Create the module with the same resource definitions, (2) Add moved blocks for each resource being moved: `moved { from = aws_instance.web, to = module.web_server.aws_instance.main }`, (3) Add the module call in the root config, (4) Run `terraform plan` — it should show only the state moves, not resource create/destroy, (5) Run `terraform apply` — Terraform updates state without touching real infrastructure, (6) After successful apply, remove the moved blocks. This is the safest approach. Without moved blocks, Terraform would destroy the old resources and create new ones in the module, causing downtime. Always test in a non-production environment first.

Scenario 17. Your RDS instance needs to be upgraded to a larger instance type with minimal downtime. How do you handle this with Terraform?

Answer:

RDS instance type changes cause a restart (a few minutes of downtime). To minimize impact: (1) Schedule during a maintenance window, (2) Change the instance class in your Terraform config: `instance_class = 'db.r6g.2xlarge'`, (3) If using Multi-AZ, AWS performs the change with a failover — the standby is upgraded first, then a failover occurs (60-120 seconds), then the old primary is upgraded. This minimizes downtime to the failover time, (4) For zero-downtime (sort of), create a read replica, promote it (requires separate Terraform state), and switch your app to it, but this is complex. For most production DBs, Multi-AZ with a maintenance window is the standard approach.

Scenario 18. You have a Terraform configuration for AWS that you need to make work across multiple AWS accounts with different configurations for each. How do you design this?

Answer:

Multi-account architecture: (1) Define one aws provider per account using aliases: `provider 'aws' { alias='dev', assume_role { role_arn = var.dev_role_arn } }, provider 'aws' { alias='prod', assume_role { role_arn = var.prod_role_arn } }`, (2) Use a central identity account where Terraform runs, assuming roles in target accounts, (3) For each resource, specify which provider: `resource 'aws_s3_bucket' 'dev_bucket' { provider = aws.dev ... }`, (4) Consider separate state files per account for isolation — a mistake in dev should not affect prod state, (5) Use Terraform workspaces or directory-per-account structure with account-specific `.tfvars` files. For large organizations, Terragrunt significantly simplifies the repetitive provider and backend config.

Scenario 19. During terraform apply, a resource fails halfway through. Terraform's state shows some resources created and some not. How do you proceed?

Answer:

Terraform applies changes sequentially (with parallelism). A mid-apply failure leaves state partially applied — some resources are in state (created successfully), others are not. Steps: (1) Read the error message carefully — it tells you which resource failed and why, (2) Fix the underlying issue (e.g., insufficient IAM permissions, quota limit, or invalid config), (3) Run terraform plan — it shows what still needs to be created (the failed and skipped resources), (4) Run terraform apply again — Terraform only tries to create resources not yet in state. It is safe to re-run. (5) If a resource is in a weird state (created but broken), use terraform taint (or -replace) to force recreation. Terraform is designed to be safely re-run after failures.

Scenario 20. You need to share VPC IDs created by one team's Terraform with another team's Terraform configuration. What is the best approach?

Answer:

Several patterns: (1) terraform_remote_state data source — Team B reads Team A's state and accesses outputs. Simple but creates coupling to Team A's state structure, (2) AWS SSM Parameter Store — Team A writes vpc_id to SSM after creating it. Team B reads it with a data source: `data 'aws_ssm_parameter' 'vpc_id' { name='/networking/vpc_id' }`. This decouples teams and works even if Team A changes their Terraform structure, (3) Terraform Cloud variable sets — shared variables available to multiple workspaces, (4) AWS Resource Share — for sharing actual resources (subnets, RAM). For most teams, SSM Parameter Store provides the best balance of simplicity, decoupling, and visibility.

Scenario 21. Your CI/CD pipeline runs terraform plan on every PR, but plans are taking 10 minutes. How do you speed this up?

Answer:

Optimization strategies: (1) Split large state files into smaller ones — a state with 20 resources plans much faster than one with 500, (2) Use -target only during development (not in CI), (3) Cache the .terraform directory between pipeline runs — provider plugins are large downloads. Use GitHub Actions cache on the .terraform/providers directory, (4) Use Terraform Cloud remote runs — they cache provider plugins and have optimized network access to cloud APIs, (5) Reduce the number of data sources — each data source makes an API call during plan, (6) Check for slow API calls using TF_LOG=trace and identify bottlenecks, (7) Increase -parallelism if the bottleneck is sequential API calls (not rate limits).

Scenario 22. A new compliance requirement says all S3 buckets must have versioning enabled. You have 200 buckets managed by Terraform. How do you add versioning efficiently?

Answer:

If all buckets are created with the same module: (1) Add the versioning configuration to the module once, (2) Bump the module version, (3) Update all callers to the new module version. That is it. If buckets are defined individually: (1) Add a variable `enable_versioning` with `default=true` to your module or use a local, (2) Add the `aws_s3_bucket_versioning` resource referencing each bucket. Use a `for_each` over your buckets map, (3) Run `terraform plan` to verify only versioning is being added, no other changes, (4) Apply. For immediate compliance without waiting for Terraform: use AWS Config auto-remediation or a script to enable versioning on all existing buckets, then codify it in Terraform to prevent future non-compliance.

Scenario 23. You want to deploy infrastructure in 5 AWS regions simultaneously. How do you structure this with Terraform?

Answer:

Multi-region deployment approach: (1) Define 5 AWS provider aliases, one per region, (2) Use a module for your infrastructure pattern, and call it 5 times with different providers: `module 'us_east' { source='./app', providers={aws=aws.us-east-1} }, module 'eu_west' { ... },` etc, (3) For independent state per region (recommended), use separate directories or workspaces per region — this limits blast radius, (4) Use `Terragrunt run-all` to apply all regions in parallel from a single command, (5) Shared global resources (IAM, Route 53 hosted zones) go in a separate configuration with a single provider. The key design principle: if `us-east-1` has a problem, it should not prevent you from managing `eu-west-1`.

Scenario 24. Your organization uses Terraform but has concerns about storing state in S3 because it contains sensitive data. What do you recommend?

Answer:

A layered security approach for state: (1) Enable S3 server-side encryption with AWS KMS — all state data is encrypted at rest with a customer-managed key, (2) Enable S3 versioning — recover from accidental corruption or deletion, (3) Enable S3 access logging — audit who accessed the state, (4) Apply strict IAM policies — only Terraform's CI role and specific users can access the state bucket, (5) Block public access on the S3 bucket — this should always be on for state buckets, (6) Consider Terraform Cloud as an alternative — state is encrypted in HashiCorp's managed service with access controls and audit logs, (7) Use `sensitive=true` on variables so values are redacted from plan output. With these controls, S3 state is production-safe.

Scenario 25. A new team member ran terraform init and got 'Error: Failed to query available provider packages'. What might be wrong?

Answer:

Common causes: (1) No internet access — the machine cannot reach registry.terraform.io. Check network/proxy settings, (2) Provider version constraint cannot be satisfied — the required version in required_providers does not exist. Check the version string for typos, (3) The .terraform.lock.hcl has checksums for a different OS — if the lock file was created on Linux and you are on Mac, re-run terraform init -upgrade, (4) Authentication issue — private registry requires a token. Run terraform login, (5) Firewall blocking the provider registry URL, (6) Air-gapped environment — use a local mirror: configure providers_installation in the CLI config to point to a local mirror instead of the public registry. Check terraform init -upgrade -reconfigure for a fresh start.

Scenario 26. You need to rotate an RDS database password managed by Terraform. How do you do this without downtime?

Answer:

Database password rotation with Terraform: (1) If using AWS Secrets Manager: update the secret value in Secrets Manager (outside Terraform or via Terraform), then update the RDS password attribute to reference the new secret. AWS RDS automatically picks up the new password. (2) If the password is in a Terraform variable: change the variable value, run terraform apply — RDS modifies the password without replacing the database. This causes a brief API call but typically no restart. (3) Use AWS Secrets Manager rotation — it handles automatic rotation on a schedule, completely outside Terraform. Configure the rotation Lambda and schedule in Terraform but let Secrets Manager handle the actual rotation. Always test password rotation in a non-production environment first.

Scenario 27. Your Terraform configuration has become a monolith — one state file with 300 resources and 30-minute apply times. How do you break it apart?

Answer:

Decomposition strategy: (1) Identify natural boundaries: networking (VPC, subnets), compute (EC2, ASG), databases (RDS, ElastiCache), shared services (IAM, Route 53), (2) Start with the lowest layer: extract networking into its own Terraform configuration with its own state, (3) Export key values as outputs (vpc_id, subnet_ids), (4) In the remaining monolith, replace direct references with terraform_remote_state data sources or SSM parameters, (5) Do this incrementally — one component at a time, using terraform state mv to move resources between states without recreating them, (6) Each extracted component should be independently deployable. The goal: each state has 20-50 resources and applies in under 5 minutes.

Scenario 28. How would you implement a canary deployment using Terraform and AWS?**Answer:**

Canary deployment with Terraform and AWS: (1) Create two Target Groups in ALB — one for stable (90% weight), one for canary (10% weight), (2) Use `aws_lb_listener_rule` with weighted target group routing in Terraform, (3) Deploy new version to canary target group (small number of EC2 instances or ECS tasks), (4) Adjust weights in Terraform (e.g., 80/20, then 50/50, then 0/100) as confidence grows, (5) Monitor metrics (error rates, latency) between weight changes, (6) If something is wrong, change weights back to 100/0 for the stable version. Terraform manages the weight configuration; the actual canary progression can be done manually or automated with a script that updates a Terraform variable and applies.

Scenario 29. You have a Terraform workspace per customer (100 customers). Adding a new resource requires 100 applies. How do you automate this?**Answer:**

Automation strategies: (1) Terraform Cloud API — write a script that loops through all workspaces and triggers a run via API: `POST /api/v2/runs` with the workspace ID. Terraform Cloud queues and executes them, (2) Atlantis — configure it to auto-apply on merge to main, triggering all workspaces that use the changed module, (3) Spacelift/env0 — these platforms have built-in stack dependency and bulk run features, (4) Custom script — use `terraform workspace list`, iterate, run `terraform apply` for each. Run in parallel with GNU `parallel` or Python threads, (5) Consider restructuring — if 100 customers use the same config, a single workspace with `for_each` over a customer map might be better than 100 separate workspaces. The right architecture depends on how different each customer's config is.

Scenario 30. During a terraform plan, you see a resource marked with '-/+' (destroy and create). You expected '+~' (update in place). Why might this happen?**Answer:**

The `-/+` symbol means the resource must be destroyed and recreated instead of updated in place. Reasons this happens: (1) You changed an attribute marked as `ForceNew` in the provider — changing an EC2 instance's AMI ID, for example, always requires a new instance, (2) The provider has determined the API does not support updating that attribute — it requires a new resource, (3) A lifecycle configuration or external factor is triggering replacement. To investigate: run `terraform plan` and look for the `'# forces replacement'` annotation next to the specific attribute. The provider documentation for each attribute says whether changes `'force a new resource'`. If you want to avoid downtime, use `create_before_destroy` lifecycle and consider the impact on dependent resources.

Scenario 31. A new security policy requires that no security groups allow unrestricted inbound access (0.0.0.0/0) except on port 443. How do you enforce and audit this with Terraform?

Answer:

Multi-layer enforcement: (1) Sentinel policy in Terraform Cloud — write a policy that parses the plan JSON and checks all `aws_security_group_rule` resources. Block any plan that adds 0.0.0.0/0 on non-443 ports, (2) tfsec or Checkov — static analysis tools that scan your .tf files for security group misconfigurations. Integrate in CI to fail PRs, (3) AWS Config rule — `aws-security-group-open-to-world` detects existing violations. Set up auto-remediation or SNS notifications, (4) Module abstraction — create a shared security group module that enforces the rule internally and cannot be bypassed, (5) For existing violations: query AWS Config or Security Hub to find non-compliant SGs, then update the Terraform configs to fix them. Use a phased approach: audit first, then enforce.

Scenario 32. You need to migrate your infrastructure from AWS to GCP. You have 50+ Terraform resources. What is your strategy?

Answer:

Cross-cloud migration strategy: (1) This is not a simple configuration change — AWS and GCP have different resource types and concepts. You cannot just swap providers, (2) Create new GCP Terraform configurations alongside existing AWS ones. Both run simultaneously during migration, (3) Migrate stateless infrastructure first: compute instances, load balancers, (4) For stateful resources (databases, storage): plan a data migration (database dump/restore, `gsutil rsync` for S3-to-GCS), (5) Use a traffic migration strategy (DNS-based or load balancer weighted routing) to gradually shift traffic from AWS to GCP, (6) Keep both environments running in parallel until GCP is stable, then destroy AWS resources with `terraform destroy`, (7) Key challenge: Terraform module reuse is limited — GCP modules are completely different from AWS modules.

Scenario 33. Your terraform apply creates an EC2 instance but the application fails to start. The terraform apply reports success. How do you add a health check?

Answer:

Terraform reports success when the AWS API accepts the resource creation — it does not wait for the application to be healthy. To add validation: (1) Use a postcondition on the instance that checks the health endpoint: `postcondition { condition = can(regex('200', data.http.healthcheck.status_code)) error_message = 'App did not start correctly' }`, (2) Use a `null_resource` with a `local-exec` provisioner that polls the URL and fails if it is not healthy within a timeout. If it fails, Terraform marks the `null_resource` as tainted, (3) Use AWS Route 53 health checks and CloudWatch alarms as external validation, (4) Implement retry logic in your application startup script (via `user_data`) that attempts to connect to dependencies before starting the app.

Scenario 34. You need to create infrastructure that is identical except for the environment name (dev/prod). You don't want to copy-paste the entire configuration. What do you do?

Answer:

DRY multi-environment approach: (1) Extract all your resources into a module, (2) Create environment-specific `.tfvars` files: `dev.tfvars` and `prod.tfvars`, each with environment-specific values (names, sizes, replica counts), (3) In a single root configuration, call the module with `var.environment` to differentiate, (4) Apply with: `terraform apply -var-file=dev.tfvars` or `terraform apply -var-file=prod.tfvars`, (5) Use separate state files for dev and prod (separate workspaces or directories), (6) For larger setups, use Terragrunt's `generate` blocks to DRY up the backend configuration too. This way, all logic is in one module, and environments differ only in their variable values. Changes to the module propagate to all environments.

Scenario 35. How would you implement infrastructure immutability with Terraform?**Answer:**

Infrastructure immutability means never modifying running resources — always replace them. Implementation: (1) Use `create_before_destroy` in the lifecycle block for all stateful resources. New resource is created first, then old one is destroyed after traffic is shifted, (2) For EC2 instances: use Launch Templates with Auto Scaling Groups. Deploy a new launch template version, do a rolling instance refresh (Terraform supports this with `aws_autoscaling_group` `refresh_strategy`), (3) Avoid in-place modifications — instead of updating a server's configuration, build a new AMI (with Packer) and deploy it, (4) Use blue-green deployment patterns, (5) Immutability reduces configuration drift because servers are never manually patched or configured — they are always built fresh from code.

You're Ready!

You have covered 150 questions across core concepts, advanced features, state management, security, enterprise patterns, and real-world scenarios. Review any weak areas, practice in the terraform console, and build small projects to solidify your knowledge. Good luck in your interview!

Key Topics to Review: Remote State • Modules • for_each • Lifecycle Rules • Sentinel • Terragrunt • Atlantis • OIDC • Provider Aliasing